統計學上課板書

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Lognormal Distribution 對數常態

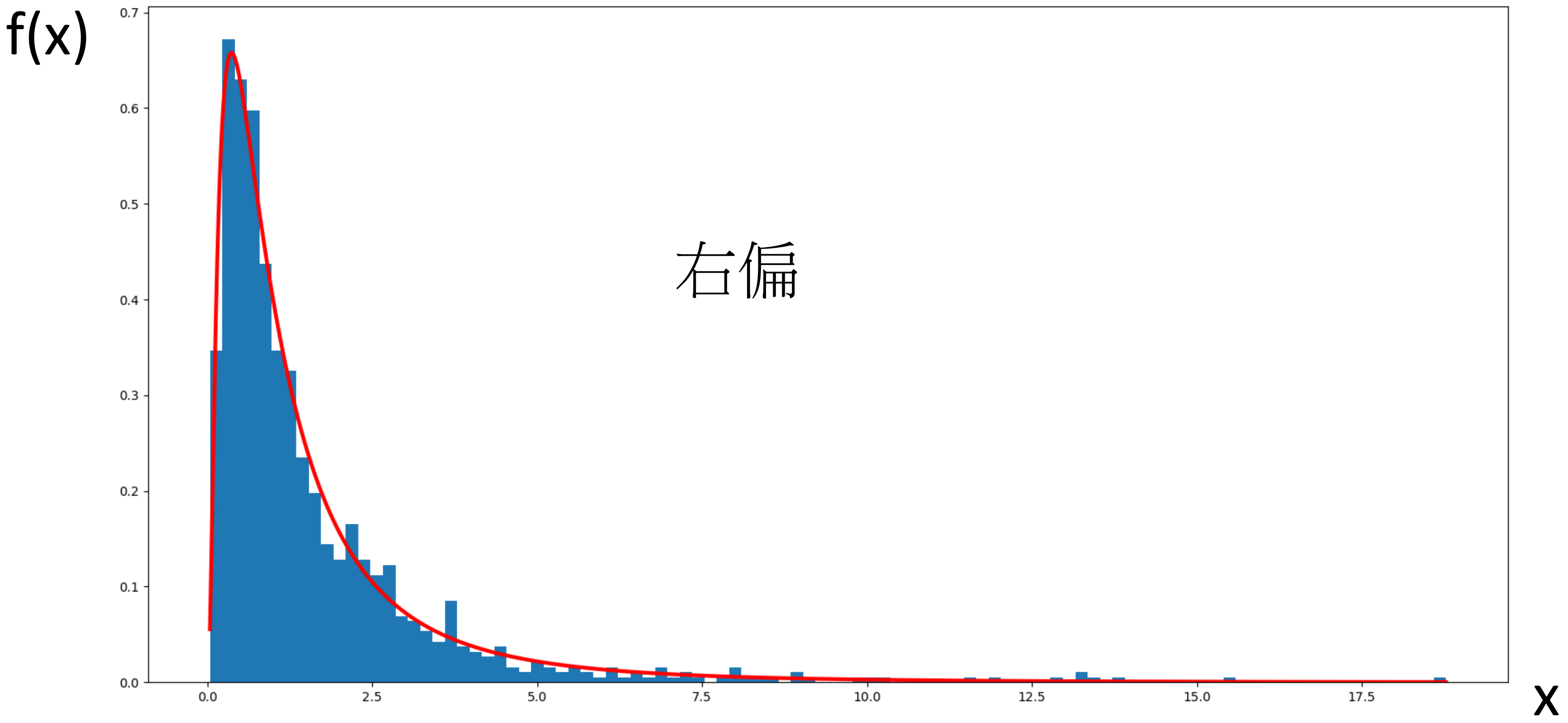
- Def： R.V. x之自然對數 $\ln x$ 呈常態分布，則x之機率分布為對數常態分布

- 若Continuous R.V. x為對數常態分布，其p.d.f f(x)

$$f(x) = \begin{cases} \frac{1}{\sqrt{2n}\beta} * x^{-1} e^{\frac{-(\ln x-\alpha)^2}{2\beta^2}} & , \ x,\beta > 0 \\ 0, & otherwise \end{cases}$$

where $\ln x$ 為x之自然對數

Demonstration of α=0, β=1



Example

- 電流的輸入與輸出比值以$^{I_o}/_{I_i}$表示之

若其自然對數$\ln(^{I_o}/_{I_i})$ 呈平均數 μ=2 &變異數 σ^2=1 之常態分布

求解 $^{I_o}/_{I_i}$ 之值，介於58～78之間之機率值為何？

$X = \ln(^{I_o}/_{I_i})$

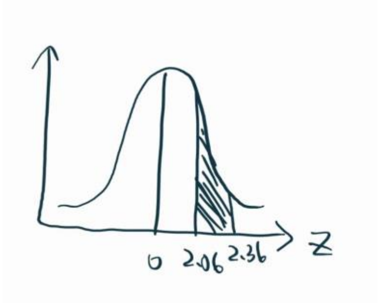
X～N(μ, σ^2) = N(2, 1)

$P(58 \le {}^{I_o}/_{I_i} \le 78)$

$=P(\ln(58) \le \ln(^{I_o}/_{I_i}) \le \ln(78))$

$=P(\frac{4.064-2}{1} < \frac{x-\mu}{\sigma} = z < \frac{4.3567-2}{1})$ 標準化：使所有尺度一樣，因此只要查表就好

$=P(2.06 < z < 2.36) = 0.0106$ (查表



Uniform Distribution 齊一分布

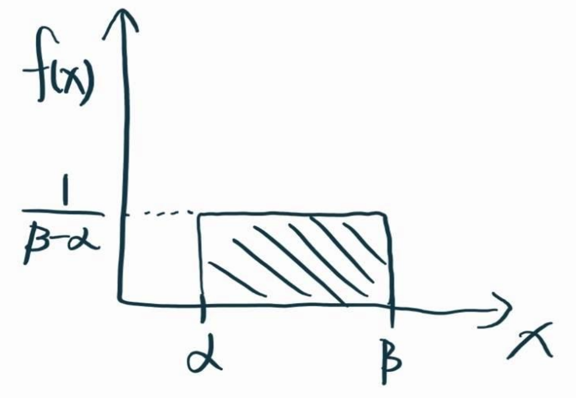
- 又稱為均等分布
- Def：Continuous R.V. X為齊一R.V.

則X在某連續區間

其有相同的機率密度

其X之p.d.f f(x) 為

$$f(x)=\begin{cases}\dfrac{1}{\beta-\alpha}, & \alpha<x<\beta\\ 0, & otherwise\end{cases}$$



Expected Value & Variance

- E(X) = $\int_{\alpha}^{\beta} xf(x)dx$ = $\int_{\alpha}^{\beta} x\frac{1}{\beta-\alpha}dx$

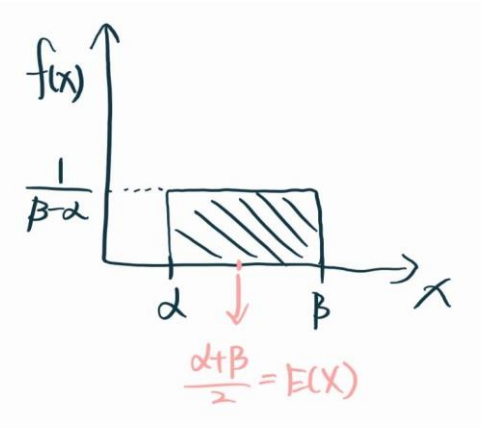
$$= \frac{1}{\beta-\alpha}\int_{\alpha}^{\beta} xdx$$

$$= \frac{1}{\beta-\alpha}\,\frac{1}{2}x^2\Big]_{\alpha}^{\beta}$$

$$= \frac{1}{\beta-\alpha}\left(\frac{\beta^2}{2} - \frac{\alpha^2}{2}\right)$$

$$= \frac{1}{\cancel{\beta-\alpha}}\left(\frac{(\beta+\alpha)\,\cancel{(\beta-\alpha)}}{2}\right) = \frac{\beta+\alpha}{2}$$

Note： β^2-α^2=(β+α) (β-α)



$\mathrm{Var}(X) = E(X^2) - \mu^2$

$$\Rightarrow \int_{\alpha}^{\beta} x^2 f(x)dx \ - \ (\frac{\alpha+\beta}{2})^2$$

$$= \int_{\alpha}^{\beta} x^2 \frac{1}{\beta-\alpha} dx \ - \ (\frac{\alpha+\beta}{2})^2$$

$$= \frac{1}{\beta-\alpha} \int_{\alpha}^{\beta} x^2 dx \ - \ (\frac{\alpha+\beta}{2})^2$$

$$\Rightarrow \frac{1}{\beta-\alpha} \frac{1}{3} x^3 \Big]_{\alpha}^{\beta} \ - \ (\frac{\alpha+\beta}{2})^2$$

此部分老師先前導公式時
有誤，請同學注意

$$= \frac{1}{\beta-\alpha} (\frac{\beta^3}{3} - \frac{\alpha^3}{3}) \ - \ (\frac{\alpha+\beta}{2})^2$$

Note：$\beta^3 - \alpha^3 = (\beta-\alpha)(\beta^2+\alpha\beta+\alpha^2)$

$$= \frac{1}{\cancel{\beta-\alpha}} \frac{\cancel{(\beta-\alpha)}\,(\beta^2+\alpha\beta+\alpha^2)}{3} \ - \ (\frac{\alpha+\beta}{2})^2$$

$$= \frac{(\beta^2+\alpha\beta+\alpha^2)}{3} - \frac{(\alpha^2+2\alpha\beta+\beta^2)}{4} = \frac{4(\beta^2+\alpha\beta+\alpha^2)-3(\alpha^2+2\alpha\beta+\beta^2)}{12} = \frac{\beta^2-2\alpha\beta+\alpha^2}{12} = \frac{(\beta-\alpha)^2}{12}$$

Example

- If X is uniformly distributed over (0, 10) $\alpha = 0, \beta = 10$

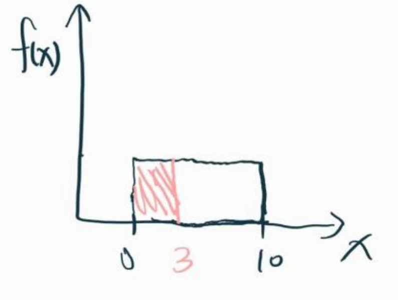
X～Uniform (0, 10)

$$\Rightarrow f(x) = \begin{cases} \frac{1}{10}, & 0 \le x \le 10 \\ 0, & otherwise \end{cases}$$

a) p(x<3)

= $\int_0^3 \frac{1}{10} dx$

=> $\frac{1}{10} x]_0^3$ = $\frac{1}{10}$ (3-0) = $\frac{3}{10}$



Example

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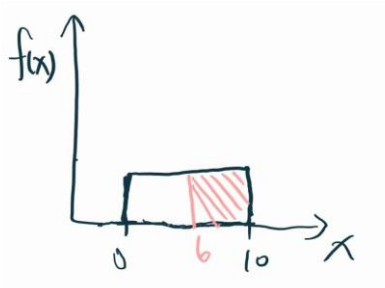
X～Uniform (0, 10)

$$\Rightarrow f(x) = \begin{cases} \frac{1}{10}, & 0 \le x \le 10 \\ 0, & otherwise \end{cases}$$

b) p(x>6)

= $\int_{6}^{10} \frac{1}{10} dx$

=> $\frac{1}{10} x \Big]_{6}^{10}$ = $\frac{1}{10}$ (10-6) = $\frac{4}{10}$



Example

- If X is uniformly distributed over (0, 10) $\alpha = 0, \beta = 10$

X～Uniform (0, 10)

$$\Rightarrow f(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10 \\ 0, & otherwise \end{cases}$$

c) p(3<x<8)

= $\int_3^8 \frac{1}{10} dx$

=> $\frac{1}{10} x]_3^8$ = $\frac{5}{10}$

Example

- 工廠生產面板，其厚度大於160mm為合格品

X：面板厚度

X～Uniform (150, 200)

$$f(x) = \begin{cases} \dfrac{1}{200-150} = \dfrac{1}{50}, & 150 < x < 200 \\ 0, & otherwise \end{cases}$$

1)面板不合格率

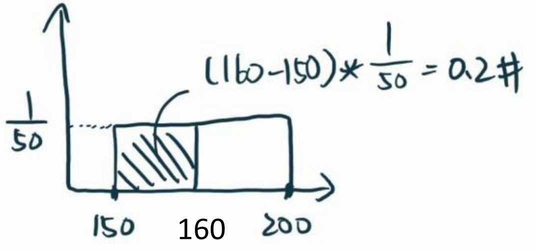
p(x≤160)

=p(150≤x≤160) Lower bound: 150 upper bound: 160 小於150的機率都是0

$= \int_{150}^{160} \frac{1}{50}\,dx$

$=> \frac{1}{50}x\Big]_{150}^{160} = \frac{1}{50}(10) = 0.2$

也可以：



*但圖要先畫對

Example

- 工廠生產面板，其厚度大於160mm為合格品

X：面板厚度

X～Uniform (150, 200)

$$f(x) = \begin{cases} \dfrac{1}{200-150} = \dfrac{1}{50}, & 150 < x < 200 \\ 0, & otherwise \end{cases}$$

2)面板厚度平均數與標準差

E(X) = $\frac{\alpha+\beta}{2}$ = $\frac{150+200}{2}$ = 175

σ(X) = $\sqrt{\frac{(\beta-\alpha)^2}{12}}$ = $\sqrt{\frac{2500}{12}}$ = 14.43376 mm

Exponential Distribution 可以想成是在探討等待時間

- Def：R.V. X為指數隨機變數 其pdf f(x)

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \,\&\, \lambda > 0 \\ 0, & otherwise \end{cases}$$

E(X) = $\frac{1}{\lambda}$

Var(X) = $\frac{1}{\lambda^2}$

其cdf F_x(t) 累積函數

F_x(t) = p(x≤t) = $\int_0^t \lambda e^{-\lambda x} dx$ = 1- $e^{-\lambda t}$

時間沒有負值

p(x>t) = $e^{-\lambda t}$

Example1

- The waiting time at a bus stop has an exponential distribution at a range of three buses arrived per hour.

What are the probabilities that the time between the arrival of buses will be?

a) Less than 5 minutes

b) At least 10 minutes

c) What is the expected waiting time between successive arrivals?

Example1

- The waiting time at a bus stop has an exponential distribution at a range of three buses arrived per hour.

What are the probabilities that the time between the arrival of buses will be?

λ = 3 buses/hr $\overset{\div 60}{=>}$ 0.05/minute

X：time between the arrival of successive buses

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & otherwise \end{cases}$$

X～Exp(λ = 0.05)

Example1

- The waiting time at a bus stop has an exponential distribution at a range of three buses arrived per hour.

What are the probabilities that the time between the arrival of buses will be?

a) Less than 5 minutes

p(x<5) = 1- $e^{-\lambda x}$ = 1- $e^{-0.05*5}$ = 1- $e^{-0.25}$ = 0.2212

Example1

- The waiting time at a bus stop has an exponential distribution at a range of three buses arrived per hour.

What are the probabilities that the time between the arrival of buses will be?

b) At least 10 minutes

p(x>10) = $e^{-\lambda x}$ = $e^{-0.05*10}$=0.6065

Example1

- The waiting time at a bus stop has an exponential distribution at a range of three buses arrived per hour.

What are the probabilities that the time between the arrival of buses will be?

c) What is the expected waiting time between successive arrivals?

$E(X) = \frac{1}{\lambda} = \frac{1}{0.05} = 20$ minutes

Example2

服從

- Service time for customers at a information desk can be modeled by an exponential distribution with a mean service time of 5 minutes.

E(X)

What is the probability that a customer service time will be more than 8 minutes?

Example2

X：service time (minute)

X～Exp(λ)

E(X) = 5 = $\frac{1}{\lambda}$ => λ = $\frac{1}{5}$ = 0.2/minute

p(x>8) = $e^{-\lambda x}$ = $e^{-0.2*0.8}$ = $e^{-1.6}$ = 0.2019

Gamma Distribution

- $f(x) = \begin{cases} \frac{1}{\beta^{\alpha}\boldsymbol{\Gamma}(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}}, & x, \alpha, \beta > 0 \\ 0, & otherwise \end{cases}$

- $\boldsymbol{\Gamma}(\alpha)$ = $\int_0^{\infty} x^{\alpha-1} e^{-x} dx$

Beta Distribution 若資料呈現微笑曲線可用此求機率

- $f(x) = \begin{cases} \dfrac{\boldsymbol{\Gamma}(\alpha+\beta)}{\boldsymbol{\Gamma}(\alpha)\boldsymbol{\Gamma}(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, & 0 < x < 1 \text{ , } \alpha, \beta > 0 \\ 0, & otherwise \end{cases}$

Weibull Distribution

- $f(x) = \begin{cases} \alpha\beta(\beta x)^{\alpha-1}e^{-\beta x^{\alpha}}, & x > 0 \\ 0, & otherwise \end{cases}$